arXiv:2606.04243v1 [cs.SI] 2 Jun 2026

Structural properties of the implicit function defined by an integral self-consistency equation

Ivan Viakhirev
ITMO University, St. Petersburg, Russian Federation
i.viakhirev@mail.ru

Abstract

We study the integral equation

$$\int_0^m \frac{\eta\,\rho(\eta)}{C-\eta}\,d\eta = 1, \qquad C > m,$$

where ρ is a C^1 probability density on $[0, M]$ vanishing polynomially at $\eta = M$. Setting $\mathcal{I}^+(m) := \lim_{C\downarrow m} \int_0^m \eta\rho(\eta)/(C-\eta)\,d\eta$ and $\Omega := \{m \in (0, M) : \mathcal{I}^+(m) > 1\}$, the displayed equation determines C implicitly as a function of m on Ω, and our object of study is the dimensionless ratio $\beta(m) := C(m)/m$.

Write $h(\eta) := \eta\rho(\eta)$. Our main result (Theorem 7) establishes openness of Ω, C^1-smoothness of β, a sign formula identifying $\beta'(m)$ with a positively-weighted integral of $dh/d\ln\eta$, transfer of monotonicity from h to β, and existence of an interior critical point of β when h is unimodal and two technical hypotheses hold.

Numerically, β has a single critical point in seven log-concave test densities (mostly Beta-type), in support of a separate uniqueness conjecture. A bimodal density that violates both unimodality and log-concavity exhibits three critical points; this shows that dropping the two hypotheses jointly admits multiple critical points, but does not separate their roles. Applications to preferential-attachment networks are mentioned only as motivation.

1 Introduction

1.1 The equation and the object of study

Let $M \in (0, \infty)$ and let ρ be a probability density on $[0, M]$. For $m \in (0, M]$ and $C > m$, define

$$F(C, m) \;:=\; \int_0^m \frac{\eta\,\rho(\eta)}{C-\eta}\,d\eta - 1. \tag{1}$$

We study the equation $F(C, m) = 0$.

When this equation has a unique solution $C(m) > m$, the ratio

$$\beta(m) \;:=\; \frac{C(m)}{m} \tag{2}$$

is the central object of this paper. The set of m for which a solution exists is characterized by the limit

$$\mathcal{I}^+(m) \;:=\; \lim_{C\downarrow m} \left(F(C, m) + 1\right) \;=\; \lim_{C\downarrow m} \int_0^m \frac{\eta\rho(\eta)}{C-\eta}d\eta \;\in\; (0, \infty], \tag{3}$$

which exists by monotone convergence (the integrand increases as $C \downarrow m$). When $\mathcal{I}^+(m) < \infty$, the limit equals the improper integral $\int_0^m \eta\rho(\eta)/(m-\eta)d\eta$.

1.2 Origin and motivation

Equations of the form (1) arise in the mean-field analysis of preferential-attachment dynamics with heterogeneous fitness; see Section 6 for references and for the limits of the analogy. In these applications it is the ratio $\beta = C/m$, rather than $C(m)$, that carries an intrinsic meaning: it is dimensionless, it is bounded below by 1 throughout the domain, and it controls the relevant degree-distribution exponent. Analytically, the substitution $\eta = mt$ in (1) turns the equation into one for β alone on the fixed interval $[0, 1]$ (Lemma 4), which is what makes the derivative formula (Theorem 7(c)) clean.

1.3 Main results

Write $h(\eta) := \eta\rho(\eta)$. Under hypotheses (D0)–(D3) on ρ (Section 2.1), Theorem 7 establishes the following five properties of β.

(a) *Domain.* $\beta(m)$ is defined exactly on the open set $\Omega := \{m \in (0, M) : \mathcal{I}^+(m) > 1\} \subset (0, M)$.

(b) *Smoothness.* $\beta \in C^1(\Omega)$ and $\beta > 1$ on Ω.

(c) *Sign formula.* For $m \in \Omega$,

$$\operatorname{sign}\big(\beta'(m)\big) \;=\; \operatorname{sign}\left(\int_0^1 \frac{1}{\beta(m) - t} \cdot \frac{dh}{d\ln\eta}\Big|_{\eta=mt}\, dt\right). \tag{4}$$

(d) *Monotonicity transfer from h to β.* On any sub-interval of Ω where h is monotone and non-constant, β is strictly monotone in the same direction.

(e) *Critical point under unimodality.* Suppose h is unimodal with mode $\eta_h^* \in (0, M)$, and let $\mathcal{J}(m)$ denote the integral on the right-hand side of (4). If there is $m^\sharp \in (\eta_h^*, M)$ with $[\eta_h^*, m^\sharp] \subset \Omega$ and $\mathcal{J}(m^\sharp) < 0$, then β has a critical point $m^\star \in (\eta_h^*, m^\sharp)$.

The content of (4) is that $\beta'(m)$ has the sign of a positive-weight average of the logarithmic derivative $dh/d\ln\eta$ over $(0, m]$: β inherits monotonicity from the elasticity of h, and a critical point of β is the location at which positive and negative contributions of $dh/d\ln\eta$, weighted by the implicit-function kernel $1/(\beta(m) - t)$, balance. Part (d) is the case where the elasticity has a definite sign throughout; part (e) is the simplest case where it changes sign once. Part (e) requires two technical hypotheses, conventionally labelled

(H1) $\mathcal{J}(m^\sharp) < 0$ for some $m^\sharp \in (\eta_h^*, M)$,

(H2) $[\eta_h^*, m^\sharp] \subset \Omega$,

which we will refer to by these names. (H1) is the substantive sign-change condition; (H2) is automatic whenever $\rho > 0$ on $(0, M)$ (in particular for log-concave h). We verify (H1) numerically for all seven log-concave test densities of Section 5.

A uniqueness conjecture. If, in addition to the hypotheses of (e), h is log-concave on $(0, M)$, we conjecture that $m^\star$ is the unique critical point of β on Ω (Conjecture 10). A bimodal example showing what happens when both hypotheses fail is discussed in Section 5.

2 Setup and preliminaries

2.1 Hypotheses on ρ

We use the following hypotheses throughout. Let $M \in (0, \infty)$ and let ρ be a function on $[0, M]$.

(D0) $\rho \geq 0$ and $\int_0^M \rho(\eta)\, d\eta = 1$.

(D1) $\rho \in C^1([0, M])$, with one-sided derivatives at the endpoints.

(D2) There exist $\alpha > 1$, $K < \infty$ and $\eta_0 \in (0, M)$ such that $\rho(\eta) \leq K(M - \eta)^{\alpha - 1}$ for all $\eta \in [\eta_0, M]$. That is, ρ vanishes polynomially at $\eta = M$ at order at least $\alpha - 1$; in particular $\rho(M) = 0$. This holds for Beta-type densities $\rho(\eta) \propto \eta^a (M - \eta)^b$ with $b \geq 1$ and fails for any density with $\rho(M) > 0$ (log-normal, exponential, Gamma); the question of extending the analysis past (D2) is taken up in Section 6.

(D3) ρ is not identically zero on any subinterval of $(0, M)$. Equivalently, $\rho > 0$ on a positive-measure subset of every subinterval of $(0, M)$.

Define $h(\eta) := \eta\rho(\eta)$. By (D1), $h \in C^1([0, M])$, with derivative $h'(\eta) = \rho(\eta) + \eta\rho'(\eta)$; by (D2), $h(M) = 0$.

2.2 Preliminary lemmas

Theorem 7 relies on five lemmas: a finiteness criterion for the limit $\mathcal{I}^+$ (Lemma 1), C^1-regularity and strict C-monotonicity of F (Lemma 2), boundary values of F in C (Lemma 3), the reparametrisation $\eta = mt$ that makes $\beta = C/m$ the only unknown (Lemma 4), and openness of Ω (Lemma 5). Of these, only Lemma 1 uses the polynomial-decay hypothesis (D2).

Lemma 1 (Finiteness criterion for $\mathcal{I}^+$)**.** *Under* (D0), (D1), (D2) *with exponent* $\alpha > 1$*:*

(i) *At the endpoint $m = M$: $\mathcal{I}^+(M) < \infty$, driven by the polynomial vanishing of ρ supplied by (D2).*

(ii) *At interior $m \in (0, M)$: $\mathcal{I}^+(m) < \infty$ iff $\rho(m) = 0$, and $\mathcal{I}^+(m) = +\infty$ otherwise. The integrability when $\rho(m) = 0$ comes from the local Lipschitz bound on ρ implied by (D1).*

Proof. (i): By monotone convergence (the integrand $\eta\rho(\eta)/(C-\eta)$ increases pointwise as $C \downarrow M$ to $\eta\rho(\eta)/(M - \eta)$),

$$\mathcal{I}^+(M) \;=\; \int_0^M \frac{\eta\rho(\eta)}{M - \eta}\, d\eta, \tag{5}$$

which is finite by the following splitting. On $[0, \eta_0]$ the integrand $\eta\rho(\eta)/(M - \eta)$ is continuous on a compact set with denominator $\geq M - \eta_0 > 0$, hence bounded. On $[\eta_0, M]$, hypothesis (D2) gives

$$\frac{\eta\rho(\eta)}{M - \eta} \;\leq\; \frac{MK(M - \eta)^{\alpha - 1}}{M - \eta} \;=\; MK(M - \eta)^{\alpha - 2}, \tag{6}$$

and $\int_{\eta_0}^M (M - \eta)^{\alpha - 2}\, d\eta < \infty$ since $\alpha > 1$. Hence $\mathcal{I}^+(M) < \infty$.

(ii), $\Leftarrow$: Assume $\rho(m) = 0$ with $m \in (0, M)$. By (D1), $\|\rho'\|_\infty < \infty$, and the mean value theorem at $\rho(m) = 0$ gives $\rho(\eta) \leq \|\rho'\|_\infty (m - \eta)$ on $[m/2, m]$. For $C > m$, $(m-\eta)/(C-\eta) \leq 1$, so on $[m/2, m]$ the integrand $\eta\rho(\eta)/(C-\eta)$ is dominated by $m\, \|\rho'\|_\infty$; on $[0, m/2]$ it is dominated by $\|\rho\|_\infty$ (because $C - \eta \geq m/2$). The resulting constant dominator is L^1, so dominated convergence as $C \downarrow m$ gives

$$\mathcal{I}^+(m) \;=\; \int_0^m \frac{\eta\rho(\eta)}{m - \eta}\, d\eta \;<\; \infty. \tag{7}$$

(ii), $\Rightarrow$: Assume $\rho(m) > 0$. By continuity, $\rho(\eta) \geq \rho(m)/2$ on $[m-\delta, m]$ for some $\delta > 0$. For C slightly greater than m,

$$\int_{m-\delta}^{m} \frac{\eta\rho(\eta)}{C-\eta}\, d\eta \;\geq\; \frac{(m-\delta)\rho(m)}{2}\, \ln\frac{C-(m-\delta)}{C-m} \;\longrightarrow\; +\infty \quad \text{as } C \downarrow m, \tag{8}$$

so $\mathcal{I}^+(m) = +\infty$. □

Remark 1 (Implication for Ω). Lemma 1(ii) gives $\mathcal{I}^+(m) = +\infty > 1$ at every m with $\rho(m) > 0$, so

$$\{m \in (0, M) : \rho(m) > 0\} \;\subset\; \Omega \;\subset\; \{m \in (0, M) : \rho(m) > 0\} \cup \{m : \rho(m) = 0\}. \tag{9}$$

By (D3), the lower set is dense in $(0, M)$. Whether M is a limit point of Ω from below depends on whether $\mathcal{I}^+(M) > 1$, a nontrivial extra condition.

Lemma 2 (Smoothness and strict monotonicity in C)**.** *Under* (D0), (D1)*:*

(i) $F \in C^1\big(\{(C, m) : 0 < m < M, C > m\}\big)$*;*

(ii) $\partial_C F(C, m) = -\int_0^m \eta\rho(\eta)/(C-\eta)^2\, d\eta \leq 0$*;*

(iii) $\partial_m F(C, m) = m\rho(m)/(C-m)$*.*

Under additionally (D3)*:* $\partial_C F(C, m) < 0$ *strictly for all* (C, m) *in the domain.*

Proof. Part (i)–(iii) under (D0), (D1) *only.* We prove C^1 on each compact set $K_{m_0,m_1,\varepsilon} := \{(C, m) : m_0 \leq m \leq m_1,\ m + \varepsilon \leq C \leq 1/\varepsilon\}$ for fixed $0 < m_0 < m_1 < M$, $\varepsilon \in (0, m_0/4)$; the open domain is exhausted by such sets. Since C^1-regularity is a local property, this suffices to conclude $F \in C^1$ on the full open domain.

Continuity of F. The integrand $\eta\rho(\eta)/(C-\eta)$ is continuous in (C, m, η) on the relevant set, and bounded by $M\|\rho\|_\infty/\varepsilon$ uniformly. Dominated convergence gives continuity.

$\partial_C F$. The pointwise derivative $-\eta\rho(\eta)/(C-\eta)^2$ is bounded by $M\|\rho\|_\infty/\varepsilon^2$, hence integrable, so differentiation under the integral sign (Folland, 1999, Thm. 2.27) applies and yields the formula for $\partial_C F$. The integrand $\eta\rho(\eta)/(C-\eta)^2$ is non-negative by (D0), so $\partial_C F \leq 0$.

$\partial_m F$. The integrand of F does not depend on m except through the upper limit; the Leibniz rule yields $\partial_m F = m\rho(m)/(C-m)$, continuous since $\rho \in C^1$ and $C - m \geq \varepsilon$.

Both partial derivatives are continuous on the compact set $K_{m_0,m_1,\varepsilon}$ by dominated convergence, and a standard result on functions of two variables (e.g. Folland, 1999, §2.5) gives $F \in C^1$ there. This establishes (i)–(iii) under (D0), (D1).

Strict negativity under (D3). Under (D3), $\rho > 0$ on an open subinterval of $(0, m)$, on which the non-negative integrand $\eta\rho(\eta)/(C-\eta)^2$ is strictly positive. The integral is therefore strictly positive, giving $\partial_C F < 0$ strictly throughout the domain. □

Lemma 3 (Boundary values)**.** *Under* (D0), (D1)*, for fixed $m \in (0, M]$:*

(i) $\lim_{C \downarrow m} F(C, m) = \mathcal{I}^+(m) - 1$ *(with $+\infty$ allowed);*

(ii) $\lim_{C \to \infty} F(C, m) = -1$.

Proof. (i): By definition of $\mathcal{I}^+$.

(ii): For $C \geq 2m$, $\eta \in [0, m]$: $C - \eta \geq m$, so $\eta\rho(\eta)/(C-\eta) \leq \rho(\eta)$, integrable. Pointwise the integrand $\to 0$. Dominated convergence gives $F \to -1$. □

Lemma 4 (Reparametrization)**.** *For $m > 0$ and $C > m$, the substitution $\eta = mt$ gives $F(C, m) + 1 = m\Phi(C/m, m)$, where*

$$\Phi(\beta, m) \;:=\; \int_0^1 \frac{t\rho(mt)}{\beta - t}\, dt, \qquad \beta > 1. \tag{10}$$

Equivalently, $F(C, m) = 0$ iff $m\Phi(\beta, m) = 1$ where $\beta = C/m$.

Proof. The substitution $\eta = mt$, $d\eta = m\,dt$ sends $\eta \in [0, m]$ to $t \in [0, 1]$, and

$$\int_0^m \frac{\eta\rho(\eta)}{C-\eta}\,d\eta \;=\; \int_0^1 \frac{(mt)\rho(mt)\cdot m}{C-mt}\,dt \;=\; m\int_0^1 \frac{t\rho(mt)}{(C/m)-t}\,dt \;=\; m\,\Phi(\beta, m). \tag{11}$$

□

Lemma 5 (Openness of Ω)**.** *Define* $\Omega := \{m \in (0, M) : \mathcal{I}^+(m) > 1\}$. *Under* (D0), (D1), (D2), (D3), Ω *is open in* $(0, M)$.

Proof. Fix $m_0 \in \Omega$.

Case $\mathcal{I}^+(m_0) < \infty$*:* By Lemma 3(i), $\lim_{C\downarrow m_0} F(C, m_0) = \mathcal{I}^+(m_0) - 1 > 0$, so by strict inequality and continuity in C there exists $C_0 > m_0$ with $F(C_0, m_0) > 0$.

Case $\mathcal{I}^+(m_0) = +\infty$*:* By Lemma 3(i) (with $+\infty$ allowed), there exists $\delta_0 > 0$ with $F(m_0 + \delta_0, m_0) > 0$. Set $C_0 := m_0 + \delta_0$; by Lemma 2, F is continuous in a neighbourhood of (C_0, m_0), so there exists $\varepsilon > 0$ with $F > 0$ on the open ball $B((C_0, m_0), \varepsilon)$. Pick $\varepsilon' < \min(\varepsilon, \delta_0/2)$ (the bound $\delta_0/2$ keeps the projection $U := \{m : |m - m_0| < \varepsilon'\}$ strictly below C_0, so $C_0 > m$ on U and the pair (C_0, m) stays in the domain of F). Then $F(C_0, m) > 0$ for all $m \in U$.

In both cases we obtain an open neighborhood $U \ni m_0$ in $(0, M)$ with $\sup U < C_0$ and $F(C_0, m) > 0$ for all $m \in U$.

For $m \in U$: $F(\cdot, m)$ is strictly decreasing in C (Lemma 2(ii) under (D3)) with $F(C_0, m) > 0$ and $F \to -1$ as $C \to \infty$. Hence $\sup_{C>m} F(C, m) \geq F(C_0, m) > 0$, which by Lemma 3(i) equals $\mathcal{I}^+(m) - 1$. Therefore $\mathcal{I}^+(m) > 1$ and $m \in \Omega$. □

3 The structural theorem

Assumption 6 (Unimodality)**.** $h(\eta) = \eta\rho(\eta)$ has a mode $\eta_h^* \in (0, M)$, in the sense that h is strictly increasing on $(0, \eta_h^*)$ and strictly decreasing on (η_h^*, M).

Define

$$\mathcal{J}(m) \;:=\; \int_0^1 \frac{1}{\beta(m)-t}\cdot\frac{dh}{d\ln\eta}\Big|_{\eta=mt} dt \;=\; m\int_0^1 \frac{t\,h'(mt)}{\beta(m)-t}\,dt, \tag{12}$$

the second equality by $dh/d\ln\eta = \eta h'(\eta)$ at $\eta = mt$, which gives $(dh/d\ln\eta)|_{\eta=mt} = mt\,h'(mt)$ and pulls the factor m outside the integral. The integrand of the second form is continuous on $[0, 1]$ and bounded by $\|h'\|_\infty/(\beta(m) - 1)$, so $\mathcal{J}(m)$ is well-defined wherever $\beta(m)$ is.

Theorem 7 (Properties of the implicit ratio β)**.** *Assume* (D0), (D1), (D2), (D3) *with exponent* $\alpha > 1$, *and set*

$$\Omega \;:=\; \{m \in (0, M) : \mathcal{I}^+(m) > 1\}. \tag{13}$$

Then:

(a) (Domain.) Ω *is open in* $(0, M)$. *For each* $m \in \Omega$, *equation* $F(C, m) = 0$ *has a unique solution* $C(m) \in (m, \infty)$. *For* $m \in (0, M) \setminus \Omega$, *no solution exists.*

(b) (Smoothness.) $\beta(m) := C(m)/m \in C^1(\Omega)$, *with* $\beta(m) > 1$ *on* Ω.

(c) (Sign formula.) *On* Ω,

$$\beta'(m) \;=\; \frac{\mathcal{J}(m)}{-m^2\,\partial_\beta\Phi(\beta(m), m)}, \qquad \operatorname{sign}\big(\beta'(m)\big) \;=\; \operatorname{sign}\big(\mathcal{J}(m)\big), \tag{14}$$

with $\partial_\beta\Phi(\beta(m), m) < 0$ *so the denominator of* (14) *is strictly positive.*

(d) (Monotonicity transfer.) *If* h *is non-decreasing and not constant on* $(0, m)$, *then* $\beta'(m) > 0$. *If* h *is non-increasing and not constant on* $(0, m)$, *then* $\beta'(m) < 0$.

(e) (Critical point under unimodality.) *Suppose Assumption 6 holds and:*

(H1) *there exists* $m^\sharp \in (\eta_h^*, M)$ *with* $\mathcal{J}(m^\sharp) < 0$*;*

(H2) $[\eta_h^*, m^\sharp] \subset \Omega$.

Then there exists $m^\star \in (\eta_h^*, m^\sharp)$ *with* $\mathcal{J}(m^\star) = 0$*, equivalently* $\beta'(m^\star) = 0$.

Beyond Theorem 7 we conjecture a uniqueness statement under log-concavity of h; see Conjecture 10 and Section 5.

Remark 2 (Reading the sign formula). The integrand of $\mathcal{J}(m)$ in either form of (12) is continuous on $[0,1]$: the denominator $\beta(m) - t \geq \beta(m) - 1 > 0$ is bounded away from zero on Ω and h' is bounded by (D1). The prefactor $1/|\partial_\beta \Phi(\beta(m), m)|$ in (14) becomes large as $m \to \partial\Omega$ (where $\beta(m) \downarrow 1$), so the implicit-function setup degenerates at the boundary of the domain.

The per-point monotonicity in part (d) lifts to a sub-interval statement:

Corollary 8 (Sub-interval monotonicity)**.** *If h is non-decreasing and not constant on* $(0, m)$ *for every* m *in a sub-interval* $J \subset \Omega$ *(in particular, if h is non-decreasing and not constant on* $(0, \sup J)$*), then* β *is strictly increasing on* J*. Symmetrically, if h is non-increasing and not constant on* $(0, m)$ *for every* $m \in J$*, then* β *is strictly decreasing on* J.

Remark 3 (On (H1)). Identifying analytical sufficient conditions on ρ that imply (H1) – for instance, via asymptotics of $\beta(m)$ as $m \to M$ – is left open; see Section 6.

Proposition 9 (Behavior of β at $\partial\Omega$)**.** *If* $m_n \in \Omega$ *with* $m_n \to m_\infty \in \partial\Omega \cap [0, M]$ *and* $\mathcal{I}^+(m_\infty) = 1$*, then* $\beta(m_n) \to 1$.

Proof. Passing to a subsequence, either $\beta(m_n) \to \beta_\infty \in [1, \infty)$ or $\beta(m_n) \to \infty$.

Case $\beta(m_n) \to \infty$. Equivalently $C_n = \beta(m_n) m_n \to \infty$. Fix $C^* \geq 2M$ large enough that $F(C^*, m_\infty) < 0$ (possible by Lemma 3(ii)). By continuity of F in m, $F(C^*, m_n) \to F(C^*, m_\infty) < 0$. Since $C_n \to \infty$, eventually $C_n > C^*$, and then $F(C_n, m_n) \leq F(C^*, m_n) < 0$ by monotonicity of F in C (Lemma 2(ii)), contradicting $F(C_n, m_n) = 0$.

Case $\beta_\infty > 1$. Then $F(\beta_\infty m_\infty, m_\infty) < \sup_{C > m_\infty} F(C, m_\infty) = \mathcal{I}^+(m_\infty) - 1 = 0$, again contradicting $F(C(m_n), m_n) = 0 \to F(\beta_\infty m_\infty, m_\infty)$. □

4 Proof of Theorem 7

The proof draws on Lemmas 1–5. Part (a) is the implicit existence statement: for each $m \in \Omega$, $F(\cdot, m)$ is strictly decreasing on (m, ∞) (Lemma 2) and crosses zero between its $+\infty$-side limit $\mathcal{I}^+(m) - 1 > 0$ (Lemma 3(i)) and its limit -1 at infinity (Lemma 3(ii)), so $C(m)$ exists and is unique. Smoothness in (b) is then the implicit-function theorem. The sign formula (c) comes from differentiating $G(\beta(m), m) \equiv 0$ in the reparametrisation (10) of Lemma 4. Parts (d) and (e) follow from (c).

4.1 Proof of (a): domain

Openness of Ω is Lemma 5.

For $m \in \Omega$: Lemma 2(ii) gives $F(\cdot, m)$ strictly decreasing on (m, ∞). Lemma 3(i) gives $\lim_{C \downarrow m} F = \mathcal{I}^+(m) - 1 > 0$. Lemma 3(ii) gives $F \to -1 < 0$. By continuity and strict monotonicity, a unique $C(m) \in (m, \infty)$ satisfies $F(C(m), m) = 0$.

For $m \in (0, M) \setminus \Omega$: $\sup_{C > m} F(C, m) = \mathcal{I}^+(m) - 1 \leq 0$. Since $F(\cdot, m)$ is strictly decreasing in C (Lemma 2(ii)), the supremum equals the limit as $C \downarrow m$ and is not attained. Hence $F(C, m) < \mathcal{I}^+(m) - 1 \leq 0$ for all $C > m$, and the equation $F(C, m) = 0$ has no solution.

4.2 Proof of (b): smoothness and bound

Fix $m_0 \in \Omega$. Since $C(m_0) > m_0$ by part (a), the point $(C(m_0), m_0)$ lies in the open domain $\{(C, m) : C > m,\ 0 < m < M\}$ where $F \in C^1$ (Lemma 2). Since Ω is open (part (a)), there is an open neighborhood $V \subset \Omega$ of m_0. By Lemma 2, $F \in C^1$ near $(C(m_0), m_0)$, and $\partial_C F < 0$ there. The implicit function theorem (see e.g. Krantz and Parks, 2002) produces a C^1 function $\widetilde{C}$ on some open $V_0 \subset V$ containing m_0, with $F(\widetilde{C}(m), m) = 0$ and $\widetilde{C}(m_0) = C(m_0)$. Uniqueness of the solution at each $m \in V_0$ (part (a)) forces $\widetilde{C} = C$ on V_0. Since m_0 is arbitrary, $C \in C^1(\Omega)$, hence $\beta = C/m \in C^1(\Omega)$.

Bound $\beta(m) > 1$: $C(m) > m$ by part (a).

4.3 Proof of (c): sign formula

We work with the reparametrized form (Lemma 4): on Ω, $G(\beta(m), m) = 0$ where $G(\beta, m) := m\Phi(\beta, m) - 1$.

Step 1: Setup. The proof of (c) uses the C^1 regularity of β established in (b); the chain rule below applies. Fix $m_0 \in \Omega$. By openness (Ω open) and part (b) (β continuous), there is a compact neighborhood $K \subset \Omega$ of m_0 on which $\beta_K := \min_{m \in K} \beta(m) > 1$ is attained. Pick $\beta_- \in (1, \beta_K)$ and $\beta_+ > \max_{m \in K} \beta(m)$.

Step 2: Partial derivatives of G. On the rectangle $[\beta_-, \beta_+] \times K$ the denominator $\beta - t$ is bounded below by $\beta_- - 1 > 0$, and ρ and h' are bounded by (D1). The pointwise β- and m-derivatives of the integrand of $G + 1$ are therefore dominated by constants ($\|\rho\|_\infty/(\beta_- - 1)^2$ and $\|h'\|_\infty/(\beta_- - 1)$ respectively), so differentiation under the integral sign applies and gives

$$\partial_\beta \Phi(\beta, m) = -\int_0^1 \frac{t\,\rho(mt)}{(\beta - t)^2}\,dt \;<\; 0, \tag{15}$$

$$\partial_m G(\beta, m) = \int_0^1 \frac{t\,h'(mt)}{\beta - t}\,dt, \tag{16}$$

both continuous in (β, m) by dominated convergence. The strict inequality $\partial_\beta \Phi < 0$ uses (D3). Hence $G \in C^1$ on $[\beta_-, \beta_+] \times K$, and $\partial_\beta G = m\,\partial_\beta \Phi < 0$.

Step 3: Identify $\partial_m G$ with $\mathcal{J}(m)/m$. The second form of (12) gives directly

$$\partial_m G(\beta(m), m) \;=\; \int_0^1 \frac{t\,h'(mt)}{\beta(m) - t}\,dt \;=\; \frac{\mathcal{J}(m)}{m}. \tag{17}$$

Step 4: Chain rule. Differentiating $G(\beta(m), m) \equiv 0$ at $(\beta(m), m)$ gives $0 = \partial_\beta G \cdot \beta'(m) + \partial_m G$, so

$$\beta'(m) \;=\; -\frac{\mathcal{J}(m)/m}{m\,\partial_\beta \Phi(\beta(m), m)} \;=\; \frac{\mathcal{J}(m)}{-m^2\,\partial_\beta \Phi(\beta(m), m)}. \tag{18}$$

The prefactor is strictly positive on Ω since $\partial_\beta \Phi < 0$, so $\operatorname{sign}(\beta'(m)) = \operatorname{sign}(\mathcal{J}(m))$.

4.4 Proof of (d): monotone case

Suppose h is non-decreasing and not constant on $(0, m)$. By (D1), $h \in C^1$, so $h' \geq 0$ everywhere on $(0, m)$ (a C^1 non-decreasing function has non-negative derivative pointwise).

If $h' \equiv 0$ on $(0, m)$, then h is constant, contradicting the non-constancy assumption. Since h is non-constant on $(0, m)$, there exist $a < b$ in $(0, m)$ with $h(a) < h(b)$; by the mean value theorem, there is $\eta_1 \in (a, b)$ with $h'(\eta_1) = (h(b) - h(a))/(b - a) > 0$. By continuity of h' (which is in C^0 since $h \in C^1$), there is an open interval $I \subset (0, m)$ containing η_1 on which $h' \geq \delta > 0$ for some δ. The set $T := \{t \in (0, 1) : mt \in I\}$ is an open subinterval of $(0, 1)$ of positive Lebesgue measure.

In $\mathcal{J}(m) = m \int_0^1 t\,h'(mt)/(\beta(m) - t)dt$ (second form of (12)):

- the prefactor $m > 0$;
- the weight $t/(\beta(m) - t) > 0$ for $t \in (0, 1]$ (since $\beta(m) > 1 \geq t$);
- $h'(mt) \geq 0$ everywhere;
- $h'(mt) \geq \delta$ on T.

Hence the integrand is ≥ 0 a.e. and strictly positive on the positive-measure set T, giving $\mathcal{J}(m) > 0$. By part (c), $\beta'(m) > 0$. The non-increasing case is symmetric.

4.5 Proof of (e): conditional critical point

Step 1: $\mathcal{J}(\eta_h^*) > 0$. By (H2), the closed interval $[\eta_h^*, m^\sharp] \subset \Omega$; in particular $\eta_h^* \in \Omega$, so by part (b) $\beta(\eta_h^*) > 1$ is well-defined and $\mathcal{J}(\eta_h^*)$ is well-defined. By Assumption 6, h is strictly increasing on $(0, \eta_h^*)$, hence non-decreasing and not constant on $(0, \eta_h^*)$. The hypothesis of part (d) (with $m = \eta_h^*$) is thus satisfied, and applying (d) at $m = \eta_h^*$ yields $\beta'(\eta_h^*) > 0$, equivalently (by (c)) $\mathcal{J}(\eta_h^*) > 0$.

Step 2: Sign change. (H1) gives $\mathcal{J}(m^\sharp) < 0$.

Step 3: Continuity of $\mathcal{J}$ on $[\eta_h^*, m^\sharp]$. By (H2) the compact interval $[\eta_h^*, m^\sharp]$ lies in Ω, and by (b) β is continuous there, so $\beta_- := \min_{[\eta_h^*, m^\sharp]} \beta(m) > 1$. On this interval the integrand of $\mathcal{J}(m) = m \int_0^1 t\, h'(mt)/(\beta(m) - t)\, dt$ is uniformly bounded by $m^\sharp \|h'\|_\infty/(\beta_- - 1)$. Dominated convergence and continuity of the prefactor m then give $\mathcal{J} \in C^0([\eta_h^*, m^\sharp])$.

Step 4: Intermediate value theorem. Since $\mathcal{J}$ is continuous on $[\eta_h^*, m^\sharp]$ with $\mathcal{J}(\eta_h^*) > 0$ and $\mathcal{J}(m^\sharp) < 0$, there exists $m^\star \in (\eta_h^*, m^\sharp)$ with $\mathcal{J}(m^\star) = 0$, hence $\beta'(m^\star) = 0$ by (c).

4.6 A uniqueness conjecture and a bimodal counterexample

Theorem 7 gives no uniqueness statement for the critical point produced in part (e). We conjecture uniqueness under strict log-concavity of h, and we examine numerically a bimodal density showing what can happen when this hypothesis fails.

Conjecture 10 (Uniqueness under log-concavity)**.** *Assume* (D0)–(D3)*, hypotheses* (H1), (H2) *of Theorem 7(e), and that $h(\eta) := \eta\rho(\eta)$ is positive on $(0, M)$ and strictly log-concave:*

$$\log h\big(\lambda x + (1-\lambda)y\big) \;>\; \lambda \log h(x) + (1-\lambda)\log h(y) \qquad (x \neq y \text{ in } (0,M),\ \lambda \in (0,1)). \quad (19)$$

Then the critical point $m^\star$ produced by Theorem 7(e) is the unique critical point of β on Ω.

Remark 4 (On the hypotheses of Conjecture 10). The positivity $h > 0$ on $(0, M)$ is stated explicitly because, without it, $h(\eta) = 0$ would make $\log h = -\infty$ and (19) trivially true. Strict log-concavity together with $h(0) = h(M) = 0$ already implies unimodality, which is therefore not listed separately. Hypothesis (H1) is not implied by log-concavity; we verify it numerically in all seven log-concave test densities of Table 1. Finally, $h > 0$ on $(0, M)$ implies $\rho > 0$ there, so Lemma 1(ii) gives $\Omega = (0, M)$ and the conjecture is a statement on the full interval.

The seven log-concave rows of Table 1 support the conjecture: in each case β' has a single sign change on the tested portion of Ω. The examples span a range of vanishing orders ($\alpha \in \{3, 4, 5\}$), polynomial and non-polynomial densities, and modes throughout $(0, M)$. The list is biased toward Beta-type densities; a systematic test of non-Beta-type log-concave densities (for instance, polynomial-times-trigonometric or polynomial-times-logarithmic) would strengthen the numerical case.

To probe what fails when log-concavity is dropped, we examine the following bimodal density.

Example 11 (Bimodal ρ violating both unimodality and log-concavity)**.** For $\sigma \in (0, 0.1]$ (we take $\sigma = 0.05$) let

$$\rho(\eta) \;\propto\; \eta\,(1-\eta)^2\left[e^{-(\eta-0.3)^2/(2\sigma^2)} + e^{-(\eta-0.7)^2/(2\sigma^2)}\right], \qquad \eta \in [0,1], \tag{20}$$

normalised on $[0, 1]$. Each factor in (20) is smooth, so $\rho \in C^\infty([0,1])$ and (D1) holds. The factor η gives $\rho(0) = 0$; the factor $(1-\eta)^2$ gives $\rho(\eta) = O((1-\eta)^2)$ near $\eta = 1$, so (D2) holds with $\alpha = 3$; (D0) and (D3) are immediate. Setting $h := \eta\rho(\eta)$,

$$h(\eta) \;\propto\; \eta^2(1-\eta)^2\left[e^{-(\eta-0.3)^2/(2\sigma^2)} + e^{-(\eta-0.7)^2/(2\sigma^2)}\right], \tag{21}$$

and both factors are invariant under $\eta \mapsto 1-\eta$, so $h(\eta) = h(1-\eta)$; differentiating this symmetry at $\eta = 1/2$ gives $h'(1/2) = 0$.

We claim h is bimodal at $\sigma = 0.05$. Since $h \in C^\infty([0,1])$ with $h(0) = h(1) = 0$ and $h > 0$ on $(0, 1)$, h attains its maximum at some interior point η_M. If $\eta_M = 1/2$, the symmetry would make $1/2$ the unique global maximum and h unimodal. This is ruled out numerically: at 50-digit precision $h(1/2) \approx 1.61 \times 10^{-3}$ while $h(0.30895) \approx 1.7249$. By symmetry $1-\eta_M$ is then a second maximum, and the critical point at $1/2$ is the local minimum between them. A direct search at the same precision finds exactly three critical points of h on $(0, 1)$,

$$\eta \;\in\; \{0.30895,\; 0.5,\; 0.69105\}. \tag{22}$$

The same picture holds at $\sigma = 0.06$ (three sign changes of $\mathcal{J}$ at $m^\star \approx 0.370, 0.552, 0.751$), while at $\sigma = 0.04$ the middle root of $\mathcal{J}$ falls below our numerical tolerance and we report three sign changes only for $\sigma \geq 0.05$.

Since $\rho > 0$ on $(0, 1)$, Lemma 1(ii) gives $\Omega = (0, 1)$ and Theorem 7(a)–(d) applies; part (e) and Conjecture 10 do not, because h is neither unimodal nor log-concave.

Numerically (row 8 of Table 1), β has three critical points in this example, in contrast with the single critical point observed under log-concavity. The example shows that dropping unimodality and log-concavity jointly admits multiple critical points, but it does not separate the roles of the two hypotheses: a unimodal but non-log-concave h producing multiple critical points of β would settle which of the two is operative, and we leave that question open (Section 6).

5 Numerical evidence

5.1 Method

For each density we solve $F(C, m) = 0$ on a uniform grid of $N = 2001$ points $m_i \in [0.1, M - 10^{-3}]$ using Brent's method (`scipy.optimize.brentq`, tolerance 10^{-12}). The integral in F is computed by adaptive quadrature (`scipy.integrate.quad`, relative tolerance 10^{-10}). We then evaluate $\mathcal{J}(m)$ by the same quadrature using the analytical formula (14) of Theorem 7(c), which avoids numerical differentiation. Sign changes of $\mathcal{J}$ between consecutive grid points are flagged and refined by Brent's method on $\mathcal{J}$ to tolerance 10^{-12}. For row 8 of Table 1 we recompute all reported critical points with `mpmath` at 50-digit precision; this reproduces the double-precision values to 8 digits.

We verified robustness of the sign-change count: doubling the grid to $N = 4001$ leaves the count unchanged and shifts each $m^\star$ by $< 10^{-5}$; replacing adaptive quadrature with 64-point Gauss–Legendre shifts $m^\star$ by $< 10^{-7}$; varying the sign-change threshold over $[10^{-10}, 10^{-6}]$ leaves the count unchanged. The lower-endpoint $m_{\min} = 0.1$ is well inside Ω for all seven log-concave densities ($\mathcal{I}^+(m_{\min}) > 10$), hence away from the boundary regime where the prefactor in (14) degenerates (Proposition 9).

Table 1: Critical-point structure of $\beta(m)$. In rows 1–7 (log-concave h satisfying (D0)–(D3)) the derivative β' has exactly one sign change on the tested portion of Ω, so each row records a single $m^\star$ and the displacement Gap $:= m^\star - \eta_h^*$ of the critical point from the mode of h. Row 8 is the bimodal density of Example 11, for which β' has three interior sign changes; the "Gap" column is left blank there. Densities are written up to normalisation; for $\eta^a(1-\eta)^b$ on $[0,1]$ the normalising constant is $1/B(a+1, b+1)$.

Density on $[0,1]$	α	η_h^*	$m^\star$	Gap
Log-concave h:				
$\eta^0(1-\eta)^2$ *(Beta)*	3	0.3333	0.4289	0.0955
$\eta^1(1-\eta)^2$	3	0.5000	0.6381	0.1381
$\eta^1(1-\eta)^3$	4	0.4000	0.5181	0.1181
$\eta^2(1-\eta)^3$	4	0.5000	0.6354	0.1354
$\eta^4(1-\eta)^4$	5	0.5556	0.6900	0.1345
$\eta(1-\eta)^2 e^{-2\eta}$	3	0.3820	0.5009	0.1190
$\cos^2(\pi\eta/2)$	3	0.4159	0.5267	0.1108
Bimodal h (Example 11, $\sigma = 0.05$):				
$\rho(\eta) \propto \eta(1-\eta)^2[e^{-(\eta-0.3)^2/(2\sigma^2)} + e^{-(\eta-0.7)^2/(2\sigma^2)}]$				
bimodal h ($\alpha = 3$)	3	0.3090, 0.6911	0.3620, 0.5455, 0.7502	—

5.2 Results

Row 8: the bimodal example. The bimodality argument of Example 11 gives two local maxima of h at $\eta = 0.30895, 0.69105$ (with $h = 1.72488$, equal by reflection symmetry) separated by a local minimum at $\eta = 0.5$ with $h = 0.00161$ (peak-to-trough ratio ≈ 1075). Running the protocol of Section 5 yields three interior sign changes of $\mathcal{J}$ on $\Omega = (0,1)$ at

$$m^\star \in \{0.36196,\ 0.54547,\ 0.75023\}, \quad \beta(m^\star) \in \{1.34002,\ 1.00051,\ 1.28195\}. \tag{23}$$

The middle root sits at $\beta \approx 1.00051$, close to the boundary $\beta = 1$ where the prefactor in (14) becomes large; it would be reasonable to suspect a numerical artefact there.

Verification of the middle root. A fine-grid diagnostic gives $\mathcal{J}(0.540) \approx -0.894$, $\mathcal{J}(0.545) \approx -0.074$, $\mathcal{J}(0.550) \approx +0.689$, stable across quadrature tolerances 10^{-10}–10^{-14}. An independent `mpmath` computation at 50-digit precision reproduces all three roots in (23) to 8 digits. The middle root is real.

Three sign changes are qualitatively consistent with h being bimodal but not formally implied by it; deriving the count of zeros of $\mathcal{J}$ from the local-extremum structure of h is open (Section 6). Parts (a)–(d) of Theorem 7 apply to row 8 directly; only part (e) and Conjecture 10 are inapplicable.

6 Discussion

6.1 Bianconi–Barabási networks and condensation

In the Bianconi–Barabási model (Bianconi and Barabási, 2001), ρ is the fitness density of nodes, m is the maximum fitness, and $C(m)$ is the normalisation in the rate equation governing the growth of node degrees; the ratio $\beta = C/m$ controls the resulting degree distribution. The threshold $\mathcal{I}^+(m) = 1$ is conjecturally related to the Bose–Einstein condensation transition in the deterministic mean-field reduction. The relationship to the stochastic model is delicate: there, m is a random order statistic, and most fitness densities of empirical interest (log-normal,

exponential) fail (D2). For rigorous probabilistic treatments of preferential-attachment models with fitness, see Borgs et al. (2007) and Dereich and Mörters (2009). Theorem 7(a) characterises when the deterministic mean-field equation admits a solution; the sign formula (c) describes how the structural ratio depends qualitatively on the upper fitness limit. A rigorous reduction of the stochastic model to (1) is beyond the scope of this paper.

6.2 Reinforcement-driven selection

Selection rules of the form $P(i) \propto x_i^{\beta}$, with x_i an accumulated score, arise in several settings, including the nonlinear preferential-attachment networks of Krapivsky and Redner (2001). When a self-consistency equation of the form (1) emerges as the mean-field reduction in such a setting, the qualitative behaviour of β is described by Theorem 7.

6.3 The (D2) restriction

The principal limitation of the present analysis is hypothesis (D2), which excludes the densities most relevant in network applications (log-normal, exponential, Gamma). A natural approach is to use the boundary regularisations $\rho^{(R,\alpha)}(\eta) = \rho(\eta)(R-\eta)^{\alpha-1}/Z_{R,\alpha}$, which satisfy (D2) for finite R, and to study the limit $R \to \infty$. Whether the limit $m_\alpha^\star := \lim_{R\to\infty} m_{R,\alpha}^\star$ exists for log-normal ρ is open.

Acknowledgements

The author thanks Kirill Borodin (Moscow Technical University of Communications and Informatics, Moscow, Russian Federation; `k.n.borodin@mtuci.ru`) for proof-reading the manuscript and for several suggestions on presentation.

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